

Hilbert Space

1.1 Orthonormal Bases

Definition 1.1.1 ► orthonormal family

A family $\{e_k\}_{k \in \Gamma}$ in an inner product space is called an **orthonormal family** if

$$\langle e_j, e_k \rangle = \begin{cases} 0 & \text{if } j \neq k, \\ 1 & \text{if } j = k \end{cases}$$

for all $j, k \in \Gamma$.

Proposition 1.1.2 ► finite orthonormal families

Suppose Ω is a finite set and $\{e_j\}_{j \in \Omega}$ is an orthonormal family in an inner product space. Then

$$\left\| \sum_{j \in \Omega} \alpha_j e_j \right\|^2 = \sum_{j \in \Omega} |\alpha_j|^2$$

for every family $\{\alpha_j\}_{j \in \Omega}$ in $\mathbb{F}$.

Definition 1.1.3 ► unordered sum; $\sum_{k \in \Gamma} f_k$

Suppose $\{f_k\}_{k \in \Gamma}$ is a family in a normed vector space V . The **unordered sum** $\sum_{k \in \Gamma} f_k$ is said to **converge** if there exists $g \in V$ such that for every $\varepsilon > 0$, there exists a finite subset Ω of Γ such that

$$\left\| g - \sum_{j \in \Omega'} f_j \right\| < \varepsilon$$

for all finite sets Ω' with $\Omega \subset \Omega' \subset \Gamma$. If this happens, we set $\sum_{k \in \Gamma} f_k = g$. If there is no such $g \in V$, then $\sum_{k \in \Gamma} f_k$ is left undefined.

Theorem 1.1.4 ► linear combinations of an orthonormal family

Suppose $\{e_k\}_{k \in \Gamma}$ is an orthonormal family in a Hilbert space V . Suppose $\{\alpha_k\}_{k \in \Gamma}$ is a family in $\mathbb{F}$. Then

- (a) the unordered sum $\sum_{k \in \Gamma} \alpha_k e_k$ converges $\iff \sum_{k \in \Gamma} |\alpha_k|^2 < \infty$.
 (b) Furthermore, if $\sum_{k \in \Gamma} \alpha_k e_k$ converges, then

$$\left\| \sum_{k \in \Gamma} \alpha_k e_k \right\|^2 = \sum_{k \in \Gamma} |\alpha_k|^2.$$

Theorem 1.1.5

Suppose $\{e_k\}_{k \in \Gamma}$ is an orthonormal family in an inner product space V and $f \in V$. Then

$$\sum_{k \in \Gamma} |\langle f, e_k \rangle|^2 \leq \|f\|^2.$$

Theorem 1.1.6 ► closure of the span of an orthonormal family

Suppose $\{e_k\}_{k \in \Gamma}$ is an orthonormal family in a Hilbert space V . Then

(a)

$$\overline{\text{span}\{e_k\}_{k \in \Gamma}} = \left\{ \sum_{k \in \Gamma} \alpha_k e_k : \{\alpha_k\}_{k \in \Gamma} \text{ is a family in } \mathbb{F} \text{ and } \sum_{k \in \Gamma} |\alpha_k|^2 < \infty \right\}.$$

(b) Furthermore,

$$f = \sum_{k \in \Gamma} \langle f, e_k \rangle e_k$$

for every $f \in \overline{\text{span}\{e_k\}_{k \in \Gamma}}$.

1.2 Parseval's Identity

Definition 1.2.1 ► orthonormal basis

An orthonormal family $\{e_k\}_{k \in \Gamma}$ in a Hilbert space V is called an **orthonormal basis** of V if

$$\overline{\text{span}\{e_k\}_{k \in \Gamma}} = V.$$

Theorem 1.2.2 ► Parseval's identity

Suppose $\{e_k\}_{k \in \Gamma}$ is an orthonormal basis of a Hilbert space V and $f, g \in V$. Then

1. $f = \sum_{k \in \Gamma} \langle f, e_k \rangle e_k$;
2. $\langle f, g \rangle = \sum_{k \in \Gamma} \langle f, e_k \rangle \overline{\langle g, e_k \rangle}$;
3. $\|f\|^2 = \sum_{k \in \Gamma} |\langle f, e_k \rangle|^2$.

Definition 1.2.3 ► separable

A normed vector space is called **separable** if it has a countable subset whose closure equals the whole space.

Theorem 1.2.4 ► existence of orthonormal bases for separable Hilbert spaces

Every separable Hilbert space has an orthonormal basis.

Theorem 1.2.5 ► orthonormal bases as maximal elements

Suppose V is a Hilbert space, $\mathcal{A}$ is the collection of all orthonormal subsets of V , and Γ is an orthonormal subset of V . Then Γ is an orthonormal basis of V if and only if Γ is a maximal element of $\mathcal{A}$.

Theorem 1.2.6 ► existence of orthonormal bases for all Hilbert spaces

Every Hilbert space has an orthonormal basis.

Theorem 1.2.7 ► Riesz Representation Theorem

Suppose φ is a bounded linear functional on a Hilbert space V and $\{e_k\}_{k \in \Gamma}$ is an orthonormal basis of V . Let

$$h = \sum_{k \in \Gamma} \overline{\varphi(e_k)} e_k.$$

Then

$$\varphi(f) = \langle f, h \rangle$$

for all $f \in V$. Furthermore, $\|\varphi\| = \left(\sum_{k \in \Gamma} |\varphi(e_k)|^2 \right)^{1/2}$.